

About R

Installation: R may be obtained free of charge from the R project Website at www.r-project.org.

How to Learn R: The R website provides extensive resources for learning R. Several free introductory guides can be found along with a list of the many books about R which are now available. I have assumed a basic knowledge of R for readers of this text but it is not necessary to learn it in detail before starting on this book. Many readers are able to pick up the language from the examples in the text with reference to the help pages. For example, for more information on the `lm` function, type within R:

```
help(lm)
```

More advanced users may find the R mailing lists and online forums such as Stack-Exchange helpful for more difficult questions.

Packages: This book uses some functions and data that are not part of base R. These are collected in *packages* which you may need to install. You can find out which packages are already installed in your *library* by:

```
library()
```

Some packages, such as MASS, come with the base installation so you will not need to install them. If you have not already done so, you will need to select a location from which to download additional packages by setting your CRAN (Comprehensive R Archive Network) mirror by:

```
chooseCRANmirror()
```

You can then install packages. You will surely need to install the `faraway` package which contains data and functions specific to this text:

```
install.packages("faraway")
```

You may also wish to install the other packages used in this text which are:

```
acepack, brglm, dispmo, dplyr, earth, effects, geepack, ggplot2,
lme4, pbkrtest, pscl, randomForest, reshape2, robust, RLRsim,
rpart.plot, rstan, sandwich, sm, SuppDists, tidyr, VGAM, wavethresh
```

All these packages are used infrequently (except for `ggplot2`), so you might delay installation until you need them. Note that you only need to install packages once on a given computer unless you upgrade the version of R. The standard GUIs for Windows and Mac provide menu-based methods for package installation and maintenance which you may find more convenient. The packages MASS, `rpart`, `mgcv`, `nnet`, `splines` and `survival` are part of the “recommended” R installation; you will have these already.

Any time you want to use these particular data or functions, you will need to load the package. For example,

```
library(faraway)
```

makes the data from this text available. If you get an error message about some data or functions not being found, it's quite possible that you have forgotten to load the necessary package. The `faraway` package is mostly data but does provide the `halfnorm`, `ilogit`, `logit` and `sumary` functions used in this text.

Customization: There are several ways to customize the look of R. I set the following options to achieve the output seen in this book:

```
options(width=70,digits=5,show.signif.stars=FALSE,scipen=2)
```

The `width=70` controls the width of the printed output. You may wish to increase this, depending on your screen or the width of your page. The `digits=5` reduces the number of digits shown when printing numbers from the default of seven. Note that this does not reduce the precision with which these numbers are internally stored. One might take this further — anything more than two or three significant digits in a displayed table is usually unnecessary and more importantly, distracting. `show.signif.stars=FALSE` prevents the printing of ugly significance stars in model summaries. `scipen=2` penalizes the use of scientific notation in reporting numbers. This notation has its uses but can make it harder to appreciate the size of a number. I have also sometimes edited the R output in the text to remove extraneous output or to improve the formatting.

Updates and Errata: The code and output shown in this book were generated under R version 3.2. R is regularly updated and improved so more recent versions may show some differences in the output. Sometimes these changes can break the code in the text. This is more likely to occur in the additional packages. Please refer to the book website at <http://people.bath.ac.uk/jjf23/ELM/> for information about such changes or errors found in the text.

Interfaces: The R website provides a GUI for both Mac and Windows installations. More functionality can be found elsewhere in ESS (which runs within the Emacs environment) and RStudio. Advanced users of R often find it more convenient to prepare R commands in a separate script before running these in R. Customizations to write R for many popular text editing programs are available.

Reproducible Research: It is important that you be able to reproduce your analysis in the future if needed. Sometimes, other people will need to verify your findings or, more likely, you yourself will need to modify or update your analysis. At a minimum you should maintain a commented set of R commands that will reproduce your analysis.

Sometimes you will want produce a document that includes the R commands, text describing the analysis and the output including graphs. This might look something like the text of this book. The `knitr` and `rmarkdown` packages provide this along with ability to use formats compatible with Microsoft Word and HTML for web display.